Breadth-First Search (BFS)

BFS uses a FIFO queue (first in, first out). It processes the oldest queued node first.

Step	Processed node	Queue after processing	Path to processed node
1	A	`[B, C]`	A
2	B	`[C, D, E]`	A → B
3	C	`[D, E, F]`	A → C
4	D	`[E, F]`	A → B → D
5	E	`[F, G]`	A → B → E
6	F	`[G]`	A → C → F
7	G	`[]`	A → B → E → G

Final BFS path: `A → B → E → G`

BFS visit order: `A, B, C, D, E, F, G`

Notice that BFS processes `F` before `G`, because `F` was already ahead of `G` in the queue.

B. Depth-First Search (DFS)

DFS uses a LIFO stack (last in, first out). The supplied code pushes successors in reverse order so the first listed successor is processed first.

Step	Processed node	Stack after processing (top at right)	Path to processed node
1	A	`[C, B]`	A
2	B	`[C, E, D]`	A → B
3	D	`[C, E]`	A → B → D
4	E	`[C, G]`	A → B → E
5	G	`[C]`	A → B → E → G

Final DFS path: `A → B → E → G`

DFS visit order: `A, B, D, E, G`

DFS stops immediately when `G` is processed. It does not process the remaining `C` entry on the stack.

Verification against the supplied code

Check	Expected	Executed result
BFS path	A → B → E → G	Match

BFS visit order	A, B, C, D, E, F, G	Match
DFS path	A → B → E → G	Match
DFS visit order	A, B, D, E, G	Match

Both traces match the supplied successor ordering and the manual's expected visit orders.

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Task 2: Depth-Limited Search (DLS) and Iterative Deepening DFS (IDDFS)

The manual defines the root `A` as depth 0. Therefore, the goal path has depth 3:

`A (0) → B (1) → E (2) → G (3)`

Limits 1 and 2 cannot reach the goal; limits 3 and 4 can. DLS must distinguish `FOUND`, `CUTOFF`, and `FAILURE`. IDDFS starts at depth 0 and repeats DLS using a fresh path-local cycle check on every iteration.

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A. DLS results

Depth limit	Status	Returned path	Explanation
1	`CUTOFF`	None	Search reaches depth 1 but cannot explore deeper.
2	`CUTOFF`	None	Search reaches depth 2; `G` is still one level deeper.

3	FOUND	A → B → E → G	Goal is reached exactly at the limit.
4	FOUND	A → B → E → G	Goal is found before the maximum depth is reached.

Why are limits 1 and 2 CUTOFF, rather than FAILURE? Because unexplored successors remain beyond the limit. FAILURE means the relevant search branches have been exhausted without finding the goal or needing a deeper cutoff.

B. IDDFS results

IDDFS starts at limit 0 and increases the depth limit until the goal is found or the requested maximum is reached.

Maximum depth	Limits attempted	Final status	Returned path
1	0, 1	CUTOFF	None
2	0, 1, 2	CUTOFF	None
3	0, 1, 2, 3	FOUND	A → B → E → G
4	0, 1, 2, 3	FOUND	A → B → E → G

For maximum depth 4, IDDFS stops at limit 3 because the goal has already been found. It does not need to run limit 4.

D. Why IDDFS repeats work

- At depth 0, it checks only `A`.
- At depth 1, it searches the first level.
- At depth 2, it revisits the shallow nodes and searches deeper.
- At depth 3, it repeats those shallow visits and finally reaches `G`.